

Problem Set: Classification

SOLUTIONS

1. (a) Discriminative and Generative classification are forms of probabilistic classification in which our learning problem becomes seeking to infer the posterior probability $p_Y(y|\mathbf{x})$.

Discriminative Classification seeks to learn $p_Y(y|\mathbf{x})$ directly.

Generative Classification uses Bayes' Theorem to express the posterior as:

$$p_Y(y|\mathbf{x}) \propto p_{\mathbf{X}}(\mathbf{x}|y)p_Y(y)$$

And then seeks to learn $p_Y(y|\mathbf{x})$ indirectly by first inferring $p_{\mathbf{X}}(\mathbf{x}|y)$ and $p_Y(y)$.

(b) Misclassification loss: $\mathcal{E}(f(\mathbf{x}), y) = \mathbb{I}[y \neq f(\mathbf{x})]$

Generalisation loss: $L(\mathcal{E}, \mathcal{D}, f) = \mathbb{E}_{\mathcal{D}}[\mathbb{I}[\mathcal{Y} \neq f(\mathcal{X})]]$

Bayes' Optimal Classifier, f^* , is defined as:

$$\begin{aligned}
f^* &= \operatorname{argmin}_{f \in \mathcal{F}} \mathbb{E}_{\mathcal{D}}[\mathbb{I}[\mathcal{Y} \neq f(\mathcal{X})]] \\
&= \operatorname{argmin}_{f \in \mathcal{F}} \sum_{y \in \{0,1\}} \int p_{\mathcal{Y}}(y|\mathbf{x}) p_{\mathcal{X}}(\mathbf{x}) \mathbb{I}[y \neq f(\mathbf{x})] d\mathbf{x} \\
&= \operatorname{argmin}_{f \in \mathcal{F}} \sum_{y \in \{0,1\}} \int p_{\mathcal{Y}}(y|\mathbf{x}) p_{\mathcal{X}}(\mathbf{x}) (f(\mathbf{x})(1-y) + (1-f(\mathbf{x}))y) d\mathbf{x} \\
&= \operatorname{argmin}_{f \in \mathcal{F}} \int (p_{\mathcal{Y}}(y=1|\mathbf{x})(1-f(\mathbf{x})) + p_{\mathcal{Y}}(y=0|\mathbf{x})f(\mathbf{x})) p_{\mathcal{X}}(\mathbf{x}) d\mathbf{x} \\
&= \operatorname{argmin}_{f \in \mathcal{F}} \int (p_{\mathcal{Y}}(y=1|\mathbf{x})(1-f(\mathbf{x})) + (1-p_{\mathcal{Y}}(y=1|\mathbf{x}))f(\mathbf{x})) p_{\mathcal{X}}(\mathbf{x}) d\mathbf{x} \\
&= \operatorname{argmin}_{f \in \mathcal{F}} \int (p_{\mathcal{Y}}(y=1|\mathbf{x}) + f(\mathbf{x})(1-2p_{\mathcal{Y}}(y=1|\mathbf{x}))) p_{\mathcal{X}}(\mathbf{x}) d\mathbf{x}
\end{aligned}$$

We need to find a function, f^* , which is optimal for all $\mathbf{x} \in \operatorname{dom}(X)$, so:

$$\begin{aligned}
f^*(\mathbf{x}) &= \operatorname{argmin}_{f(\mathbf{x})} (p_{\mathcal{Y}}(y=1|\mathbf{x}) + f(\mathbf{x})(1-2p_{\mathcal{Y}}(y=1|\mathbf{x}))) \\
&= \operatorname{argmin}_{f(\mathbf{x})} M(\mathbf{x})
\end{aligned}$$

This is a discrete optimisation problem:

$$\begin{aligned}
\text{if: } f^*(\mathbf{x}) = 1 \quad &\text{then: } \min_{f(\mathbf{x})} M(\mathbf{x}) = 1 - p_{\mathcal{Y}}(y=1|\mathbf{x}) \\
\text{if: } f^*(\mathbf{x}) = 0 \quad &\text{then: } \min_{f(\mathbf{x})} M(\mathbf{x}) = p_{\mathcal{Y}}(y=1|\mathbf{x})
\end{aligned}$$

This means that if $f^*(\mathbf{x}) = 1$, optimality implies:

$$\begin{aligned}
1 - p_{\mathcal{Y}}(y=1|\mathbf{x}) &\leq p_{\mathcal{Y}}(y=1|\mathbf{x}) \\
p_{\mathcal{Y}}(y=1|\mathbf{x}) &\geq 0.5
\end{aligned}$$

So the generalisation minimiser for the Misclassification Loss can be specified entirely in term of the posterior distribution:

$$f^*(\mathbf{x}) = \begin{cases} 1 & \text{if } p_{\mathcal{Y}}(y=1|\mathbf{x}) \geq 0.5 \\ 0 & \text{if } p_{\mathcal{Y}}(y=1|\mathbf{x}) < 0.5 \end{cases}$$

This is the **Bayes' Optimal Classifier**

(c) The discriminant boundary satisfies:

$$p_Y(y = 1|\mathbf{x}) = p_Y(y = 0|\mathbf{x})$$

So, using Bayes' Rule:

$$p_Y(y = 1)p_{\mathbf{X}}(\mathbf{x}|y = 1) = p_Y(y = 0)p_{\mathbf{X}}(\mathbf{x}|y = 0)$$

Because of Bernoulli assumption:

$$\begin{aligned} p_Y(y = 1) &= \theta \\ p_Y(y = 0) &= 1 - \theta \end{aligned}$$

Because of conditional Gaussian assumption:

$$\begin{aligned} p_{\mathbf{X}}(\mathbf{x}|y = 1) &= \frac{1}{(2\pi)^{m/2}} \frac{1}{|\Sigma_1|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_1)^T \Sigma_1^{-1}(\mathbf{x} - \boldsymbol{\mu}_1)\right) \\ p_{\mathbf{X}}(\mathbf{x}|y = 0) &= \frac{1}{(2\pi)^{m/2}} \frac{1}{|\Sigma_0|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_0)^T \Sigma_0^{-1}(\mathbf{x} - \boldsymbol{\mu}_0)\right) \end{aligned}$$

Thus:

$$\begin{aligned} \theta \frac{1}{(2\pi)^{m/2}} \frac{1}{|\Sigma_1|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_1)^T \Sigma_1^{-1}(\mathbf{x} - \boldsymbol{\mu}_1)\right) &= (1-\theta) \frac{1}{(2\pi)^{m/2}} \frac{1}{|\Sigma_0|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_0)^T \Sigma_0^{-1}(\mathbf{x} - \boldsymbol{\mu}_0)\right) \\ \Rightarrow \frac{\theta}{1-\theta} \frac{|\Sigma_0|^{1/2}}{|\Sigma_1|^{1/2}} &= \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_0)^T \Sigma_0^{-1}(\mathbf{x} - \boldsymbol{\mu}_0) + \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_1)^T \Sigma_1^{-1}(\mathbf{x} - \boldsymbol{\mu}_1)\right) \\ \Rightarrow \ln\left(\frac{\theta}{1-\theta}\right) + \ln\left(\frac{|\Sigma_0|^{1/2}}{|\Sigma_1|^{1/2}}\right) &= -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_0)^T \Sigma_0^{-1}(\mathbf{x} - \boldsymbol{\mu}_0) + \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_1)^T \Sigma_1^{-1}(\mathbf{x} - \boldsymbol{\mu}_1) \\ &= -\frac{1}{2}\left(\mathbf{x}^T \Sigma_0^{-1} \mathbf{x} - 2\boldsymbol{\mu}_0^T \Sigma_0^{-1} \mathbf{x} + \boldsymbol{\mu}_0^T \Sigma_0^{-1} \boldsymbol{\mu}_0 - \mathbf{x}^T \Sigma_1^{-1} \mathbf{x} + 2\boldsymbol{\mu}_1^T \Sigma_1^{-1} \mathbf{x} - \boldsymbol{\mu}_1^T \Sigma_1^{-1} \boldsymbol{\mu}_1\right) \\ \Rightarrow \frac{1}{2} \mathbf{x}^T (\Sigma_1^{-1} - \Sigma_0^{-1}) \mathbf{x} + (\boldsymbol{\mu}_0^T \Sigma_0^{-1} - \boldsymbol{\mu}_1^T \Sigma_1^{-1}) \mathbf{x} + \frac{1}{2} (\boldsymbol{\mu}_1^T \Sigma_1^{-1} \boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^T \Sigma_0^{-1} \boldsymbol{\mu}_0) &+ \ln\left(\frac{1-\theta}{\theta}\right) + \ln\left(\frac{|\Sigma_1|^{1/2}}{|\Sigma_0|^{1/2}}\right) = 0 \end{aligned}$$

Since $\Sigma_0 = \Sigma_1$:

$$\begin{aligned} \Rightarrow (\boldsymbol{\mu}_0^T - \boldsymbol{\mu}_1^T) \Sigma_0^{-1} \mathbf{x} + \frac{1}{2} (\boldsymbol{\mu}_1^T \Sigma_0^{-1} \boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^T \Sigma_0^{-1} \boldsymbol{\mu}_0) + \ln\left(\frac{1-\theta}{\theta}\right) &= 0 \\ \Rightarrow \mathbf{w} \cdot \mathbf{x} + b &= 0 \end{aligned}$$

Where:

$$\begin{aligned} \mathbf{w} &= \Sigma_0^{-1}(\boldsymbol{\mu}_0 - \boldsymbol{\mu}_1) \\ b &= \ln\left(\frac{1-\theta}{\theta}\right) + \frac{1}{2} (\boldsymbol{\mu}_1^T \Sigma_0^{-1} \boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^T \Sigma_0^{-1} \boldsymbol{\mu}_0) \end{aligned}$$

(d) This describes a linear hyperplane or linear discriminant

- (e) We proceed in a similar way to part (c), but now we no longer assume that $\Sigma_0 = \Sigma_1$, and we end up with:

$$\mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{w} \cdot \mathbf{x} + b = 0$$

Where:

$$\mathbf{A} = \frac{1}{2}(\Sigma_1^{-1} - \Sigma_0^{-1})$$

$$\mathbf{w} = (\Sigma_0^{-1} \boldsymbol{\mu}_0 - \Sigma_1^{-1} \boldsymbol{\mu}_1)$$

$$b = \ln\left(\frac{1-\theta}{\theta}\right) + \frac{1}{2}(\boldsymbol{\mu}_1^T \Sigma_1^{-1} \boldsymbol{\mu}_1 - \boldsymbol{\mu}_0^T \Sigma_0^{-1} \boldsymbol{\mu}_0) + \ln\left(\frac{|\Sigma_1|^{1/2}}{|\Sigma_0|^{1/2}}\right)$$

This describes a quadratic hypersurface, or quadratic discriminant

(f) In Naive Bayes we assume conditional independence amongst the attributes.

For Gaussian class conditional densities this is equivalent to assuming that Σ_0, Σ_1 are diagonal matrices.

So the approach suggested in (c) represents a richer model, with more parameters to learn, but with fewer assumptions.

2. (a) Discriminative and Generative classification are forms of probabilistic classification in which our learning problem becomes seeking to infer the posterior probability $p_Y(y|\mathbf{x})$.

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Generative Classification uses Bayes' Theorem to express the posterior as:

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And then seeks to learn $p_Y(y|\mathbf{x})$ indirectly by first inferring $p_{\mathbf{X}}(\mathbf{x}|y)$ and $p_Y(y)$.

(b) Assume $\mathbf{x} = [1, x_1, \dots, x_m]^T$, i.e. the bias term is present in the attribute vector.

$$y^* = \mathbf{w} \cdot \mathbf{x} + \varepsilon$$

Where $\mathcal{Y}^*$ is a latent variable with outcome y^* , such that $y^* \sim \text{Logistic}(\mathbf{w} \cdot \mathbf{x}, 1)$

We assume the following classification model:

$$\begin{aligned} y^{(i)} &= 1 & \text{if } y^{*(i)} &\geq 0 \\ y^{(i)} &= 0 & \text{if } y^{*(i)} < 0 \end{aligned}$$

Then:

$$\begin{aligned} p_{\mathcal{Y}}(y^{(i)=1} | \mathbf{x}^{(i)}) &= \mathbb{P}(y^* \geq 0 | \mathbf{x}^{(i)}) \\ &= \mathbb{P}(\mathbf{w} \cdot \mathbf{x}^{(i)} + \varepsilon^{(i)} \geq 0) \\ &= \mathbb{P}(\varepsilon^{(i)} \geq -\mathbf{w} \cdot \mathbf{x}^{(i)}) \\ &= 1 - \mathbb{P}(\varepsilon^{(i)} < -\mathbf{w} \cdot \mathbf{x}^{(i)}) \\ &= 1 - \frac{1}{1 + \exp(\mathbf{w} \cdot \mathbf{x}^{(i)})} && \text{by the cdf property of the Logistic distribution} \\ &= \frac{\exp(\mathbf{w} \cdot \mathbf{x}^{(i)})}{1 + \exp(\mathbf{w} \cdot \mathbf{x}^{(i)})} \\ &= \frac{1}{1 + \exp(-\mathbf{w} \cdot \mathbf{x}^{(i)})} \end{aligned}$$

(c) We perform a similar derivation to the one in (a), with the following updates:

$$y^* \sim \text{Logistic}(\mathbf{w} \cdot \mathbf{x} + a, b)$$

Then:

$$\begin{aligned} p_{\mathcal{Y}}(y^{(i)} = 1 | \mathbf{x}^{(i)}) &= 1 - \frac{1}{1 + \exp((\mathbf{w} \cdot \mathbf{x}^{(i)} + a) / b)} \\ &= \frac{\exp((\mathbf{w} \cdot \mathbf{x}^{(i)} + a) / b)}{1 + \exp((\mathbf{w} \cdot \mathbf{x}^{(i)} + a) / b)} \\ &= \frac{1}{1 + \exp(-(\mathbf{w} \cdot \mathbf{x}^{(i)} + a) / b)} \\ &= \frac{1}{1 + \exp(-\tilde{\mathbf{w}} \cdot \mathbf{x}^{(i)})} \end{aligned}$$

Here:

$$\tilde{w}_i = \frac{w_i}{b} \quad \forall i \neq 0$$

$$\tilde{w}_0 = \frac{w_0}{b} + \frac{a}{b}$$

Thus instead of learning $\mathbf{w}$, we must now learn $\tilde{\mathbf{w}}$. But this amounts to just re-naming the variables!

So there is no effect from changing the values of a and b .

(d) Assume:

$$y^* = \mathbf{w} \cdot \mathbf{x} + \varepsilon$$

Where $\mathcal{Y}^*$ is a latent variable with outcome y^* , such that $y^* \sim \mathcal{N}(\mathbf{w} \cdot \mathbf{x}, 1)$

We assume the following classification model:

$$\begin{aligned} y^{(i)} &= 1 & \text{if } y^{*(i)} &\geq 0 \\ y^{(i)} &= 0 & \text{if } y^{*(i)} < 0 \end{aligned}$$

Then:

$$\begin{aligned} p_{\mathcal{Y}}(y^{(i)} = 1 | \mathbf{x}^{(i)}) &= \mathbb{P}(y^* \geq 0 | \mathbf{x}^{(i)}) \\ &= \mathbb{P}(\mathbf{w} \cdot \mathbf{x}^{(i)} + \varepsilon^{(i)} \geq 0) \\ &= \mathbb{P}(\varepsilon^{(i)} \geq -\mathbf{w} \cdot \mathbf{x}^{(i)}) \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\mathbf{w} \cdot \mathbf{x}^{(i)}}^{\infty} \exp\left(-\frac{t^2}{2}\right) dt && \text{by the cdf property of the Gaussian distribution} \end{aligned}$$

(e) Probit regression is more sensitive to outliers than Logistic regression.

Why?

In logistic regression as $x \rightarrow \infty$ the logistic sigmoid decays like $\exp(-x)$

In probit regression as $x \rightarrow \infty$ the activation function decays like $\exp(-x^2)$

(f) If $K = 2$:

$$\begin{aligned} p_{\mathcal{Y}}(y = 1|\mathbf{x}) &= \frac{\exp(\mathbf{w}_1 \cdot \mathbf{x})}{\exp(\mathbf{w}_1 \cdot \mathbf{x}) + \exp(\mathbf{w}_2 \cdot \mathbf{x})} \\ &= \frac{1}{1 + \exp((\mathbf{w}_2 - \mathbf{w}_1) \cdot \mathbf{x})} \\ &= \frac{1}{1 + \exp(\mathbf{w} \cdot \mathbf{x})} \end{aligned}$$

Where: $\mathbf{w} = \mathbf{w}_2 - \mathbf{w}_1$

And this is just Logistic Regression

(g) We have a class boundary between two classes, j and k , when:

$$\begin{aligned} p_{\mathcal{Y}}(y = j|\mathbf{x}) &= p_{\mathcal{Y}}(y = k|\mathbf{x}) \\ \frac{\exp(\mathbf{w}_j \cdot \mathbf{x})}{\sum_{i=1}^K \exp(\mathbf{w}_i \cdot \mathbf{x})} &= \frac{\exp(\mathbf{w}_k \cdot \mathbf{x})}{\sum_{i=1}^K \exp(\mathbf{w}_i \cdot \mathbf{x})} \\ \exp(\mathbf{w}_j \cdot \mathbf{x}) &= \exp(\mathbf{w}_k \cdot \mathbf{x}) \\ \mathbf{w}_j \cdot \mathbf{x} &= \mathbf{w}_k \cdot \mathbf{x} \\ (\mathbf{w}_j - \mathbf{w}_k) \cdot \mathbf{x} &= 0 \end{aligned}$$

This is the equation for a linear hyperplane, thus the discriminant boundaries are linear.